

Phishing Email Detection & Awareness System

Future Interns — Cyber Security Task 2

1. Objective

The objective of this project is to analyze phishing email samples, identify phishing indicators, examine available email-header and URL evidence, classify the emails by risk, and provide practical phishing-awareness guidance for employees.

2. Investigation Methodology

The investigation followed these steps:

1. Collect publicly documented email samples.
2. Inspect the visible sender, subject, and message content.
3. Examine available raw email headers.
4. Review sender domains and mail-delivery infrastructure.
5. Inspect URLs without opening suspicious destinations.
6. Identify social-engineering and technical indicators.
7. Classify each sample based on the combined evidence.
8. Document limitations where technical evidence was unavailable.
9. Create employee awareness and prevention guidance.

3. Sample 01 — Account Verification Phishing

Source: Future Interns — Cyber Security Task 2

Classification: PHISHING

Observed Indicators

- Sender is presented only as "Security Team".
- Generic greeting: "Dear User".
- Message claims suspicious account activity.
- Recipient is asked to verify account details immediately.
- The message creates a 24-hour deadline.
- Permanent account lock is threatened.
- The verification URL uses a suspicious domain: `http://secure-account-verify[.]com`
- The request involves a potentially sensitive account action.

Link Analysis

The URL uses HTTP and a domain that does not identify the legitimate organization being claimed in the message.

The URL was documented in defanged form and was not opened.

Header Analysis

Raw email headers were not provided with this sample.

Therefore, SPF, DKIM, DMARC, Return-Path, Received headers, and sending infrastructure cannot be verified.

Assessment

The combination of generic addressing, urgency, account-lock threats, a suspicious verification request, and a suspicious domain supports the classification of this message as phishing.

4. Sample 02 — Microsoft Account Security Phishing

Source: Publicly documented phishing-analysis sample.

Raw evidence: samples/Sample_02.eml

Classification: PHISHING

Sender Information

From:

"Microsoft Account Team" <noreply@microsoftonline-verify.com>

Return-Path:

noreply@microsoftonline-verify.com

The sender domain shown in the email is microsoftonline-verify.com, which is different from the official microsoft.com domain.

Authentication Results

The raw headers contain the following results:

- SPF: FAIL
- DKIM: FAIL
- DMARC: FAIL

These authentication failures provide technical evidence that requires additional investigation and, together with the other indicators, support the phishing classification.

Sending Infrastructure

The Received header shows:

Sending host:

mail.microsoftonline-verify.com

Underlying host shown in the header:

vps-291847.contabo.net

Sending IP:

178.238.225.91

These values are recorded directly from the raw email header. No claim is made about ownership of the IP address beyond what is shown in the sample.

Message Content

The email presents itself as a Microsoft account security alert and contains:

- "Unusual sign-in activity" warning.
- Claimed country/region: Russia.
- Claimed sign-in IP: 91.234.99.42.
- Claimed platform: Windows 10.

- Claimed browser: Chrome 120.0.
- Request to secure the account immediately.
- Microsoft-related branding.
- Mandatory-service language.
- A "Review recent activity" action link.

Important: 91.234.99.42 is the IP presented in the email body as the alleged sign-in IP. It is different from the sending IP 178.238.225.91 found in the email headers.

URL Analysis

Observed URL:

<https://bit.ly/3vF9xKz>

The URL uses the Bitly shortening service. The visible shortened URL does not reveal the final destination from the link text alone.

The URL was extracted from the raw email source but was **not opened or visited**. Therefore, the final redirect destination is not claimed.

Assessment

The combination of Microsoft impersonation, a non-Microsoft sender domain, SPF/DKIM/DMARC failures, suspicious sending infrastructure, urgency, Microsoft branding, and a shortened action URL supports the classification of this message as phishing.

5. Comparison of Observed Evidence

Evidence	Sample 01	Sample 02
Generic greeting	Yes	Not the primary indicator
Urgency	Yes	Yes
Account-security theme	Yes	Yes
Suspicious sender/domain	Yes	Yes
Suspicious URL	Yes	Yes
Raw headers available	No	Yes
SPF result	Not available	FAIL
DKIM result	Not available	FAIL
DMARC result	Not available	FAIL
Sending infrastructure	Not available	Available
URL destination visited	No	No

Sample 01 was primarily assessed using visible email content and URL indicators.

Sample 02 provided additional technical evidence through its raw email headers.

6. Employee Awareness Guidance

Phishing emails attempt to persuade recipients to click links, disclose credentials, open unsafe attachments, or perform actions that benefit an attacker.

How Employees Can Identify Suspicious Emails

1. Check the complete sender address.

Do not rely only on the display name.

2. Look for urgency and pressure.

Account-lock warnings, deadlines, threats, and unexpected security alerts should be verified before action.

3. Inspect links carefully.

Be cautious with unfamiliar domains, shortened URLs, and links that do not clearly correspond to the claimed organization.

4. Watch for sensitive-information requests.

Be cautious when an unexpected email requests passwords, OTPs, authentication codes, payment details, or account information.

5. Use authentication evidence when available.

SPF, DKIM, and DMARC results can help during technical email analysis.

6. Verify independently.

Instead of using an unexpected email link, open the organization's known official website manually or contact a trusted support channel.

Employee Do's

- Verify the sender's complete email address.
- Check suspicious links before clicking.
- Treat unexpected urgent requests cautiously.
- Report suspected phishing through the approved process.
- Verify account or payment requests using a trusted channel.
- Contact IT/security when unsure.

Employee Don'ts

- Do not click suspicious links.
- Do not enter passwords through unexpected email links.
- Do not share OTPs or authentication codes.
- Do not open unexpected attachments.
- Do not rely only on logos or display names.
- Do not allow urgency to replace verification.

If You Clicked a Suspicious Link

1. Stop entering information.
2. Close the suspicious webpage.
3. Inform the organization's IT/security team.
4. If credentials were entered, follow the organization's password-reset and incident-response procedure.
5. Report the original email.

7. Key Lessons

A phishing email does not need to contain a single obvious indicator. Multiple small indicators can become significant when considered together.

Important checks include:

- Sender identity
- Sender domain
- Reply-To and Return-Path
- Received headers
- SPF
- DKIM
- DMARC
- URL structure
- Urgency and social engineering
- Requests for sensitive actions
- Branding and impersonation

Technical evidence should be combined with message-content analysis rather than relying on a single signal.

8. Safety and Handling

Suspicious URLs were not opened or visited during this investigation.

URLs were documented in defanged form where appropriate to reduce the risk of accidental interaction.

No credentials or personal information were entered into suspicious websites.

The original Sample 02 .eml file is preserved as evidence in:

`samples/Sample_02.eml`

9. Limitations

- Sample 01 did not include raw email headers, so its authentication and delivery infrastructure could not be verified.
- Sample 02's shortened URL was not opened, so its final destination was not verified.
- The technical observations in this report are limited to evidence available in the supplied/publicly documented samples.
- No unsupported ownership claim is made about the sending IP address.

10. Evidence Included in This Repository

Sample 01

- `samples/Sample_01.txt`
- `analysis/Sample_01_Analysis.txt`
- `header-analysis/Sample_01_Header_Analysis.txt`
- `link-analysis/Sample_01_Link_Analysis.txt`
- `screenshots/Sample_01/01_email_evidence.png`

Sample 02

- `samples/Sample_02.eml`
- `samples/Sample_02.txt`
- `samples/Sample_02_Body_Extract.txt`
- `analysis/Sample_02_Analysis.txt`
- `header-analysis/Sample_02_Header_Analysis.txt`
- `link-analysis/Sample_02_Link_Analysis.txt`
- `screenshots/Sample_02/01_raw_headers.png`
- `screenshots/Sample_02/02_authentication_results.png`
- `screenshots/Sample_02/03_sender_infrastructure.png`
- `screenshots/Sample_02/04_email_body.png`
- `screenshots/Sample_02/05_url_evidence.png`

11. References

Future Interns — Cyber Security Task 2

<https://futureinterns.com/cyber-security-task-2-2026/>

Sample 02 public source

<https://github.com/jacobdcook/Phishing-Analysis-Lab>

Raw Sample 02

<https://raw.githubusercontent.com/jacobdcook/Phishing-Analysis-Lab/main/samples/sample2.eml>

12. Conclusion

This project demonstrates a practical phishing-email investigation workflow using both visible email indicators and available technical evidence.

Sample 01 demonstrates content-based phishing analysis where raw headers were unavailable.

Sample 02 demonstrates a deeper investigation using raw headers, authentication results, sender infrastructure, message content, and URL analysis.

The investigation emphasizes safe handling of suspicious emails and independent verification before users take security-sensitive actions.